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Supply Chain Scheduling

 Springer

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*This book is dedicated to our
families and friends.*

Foreword

Scheduling of activities are ubiquitous processes. Scheduling takes place on a regular basis in many application areas at multiple levels, at operational levels as well as at strategic levels. I have been interested in the general area of scheduling for more than four decades and have been continuously amazed by the variety of application domains in which these processes play a critical role. One of the more important application areas is the scheduling that takes place in supply chains, which is the topic that is being covered in this book. The importance of this application area has become even more to the forefront in the pandemic period of 2020–2021, during which many global supply chains have been hurt considerably, requiring significant amounts of rescheduling.

I have known both Zhi-Long Chen and Nick Hall for more than three decades, and at every conference that we attend, either in the USA, in China, or anywhere else, we have been looking forward to our discussions on the topics we are working on. Zhi-Long and Nick have been working on the area of supply chain scheduling already for quite a while, and this monograph is the culmination of their and other people's work in this area.

While classical scheduling has been studied since the mid-1950s, the integration of scheduling with the coordination and competition issues that arise ubiquitously in supply chains dates back about 20 years. Some of the foundational works on this topic were written by the two authors of this book. This work has become highly influential within the operations research community. At the time of writing, an online search on the book title reveals 3300 published works. To my knowledge, however, this book represents the first attempt to synthesize the main research directions within supply chain scheduling into a single source.

Supply Chain Scheduling addresses the coordination and competition of scheduling and other operational decisions in supply chains from two perspectives. The first perspective is that of a single decision maker faced with multiple types of operational decisions—for example, production and distribution, or production and pricing. The second perspective is that of multiple self-interested decision makers whose decisions collectively determine the performance of a supply chain—for example, a supplier and a manufacturer, or multiple suppliers in an assembly system.

The problems studied are identified as centralized under the first perspective, and decentralized under the second one. Each perspective addresses a wide variety of applications and theoretical issues. A consistent focus of the book is the value added by the coordinated decision-making of scheduling and other functions, relative to local or even decentralized decision-making for different parts of a problem.

The book consists of two parts. Part I consists of Chaps. 3 through 6. These chapters study centralized problems in supply chain scheduling. Chapters 3 and 4 address the integration of production scheduling and outbound distribution decisions in both offline and online settings. Chapter 5 studies the coordination of production and pricing decisions, and Chap. 6 discusses the joint consideration of subcontracting and scheduling decisions. A wide variety of optimization methodologies are employed in Part I, including dynamic programming, integer optimization, branch-and-bound, heuristic design, and worst-case and asymptotic performance analysis.

Part II consists of Chaps. 7 through 9. Chapter 7 uses classical scheduling techniques and dynamic programming to identify the cost of conflict when individual players make self-interested decisions in supply chains. The works discussed in this chapter have been strongly motivational for the development of supply chain scheduling research. Chapter 8 views supply chain scheduling problems as cooperative games, and within that context studies application to sequencing and scheduling games, as well as capacity allocation and subcontracting problems. Chapter 9 uses the perspective of noncooperative games, and studies applications with complete information as well as others where some information is private.

This monograph will undoubtedly serve as a solid basis for future research on this important topic. I feel privileged to be allowed to write a foreword to this book and put this area in a proper perspective. I am confident that this text will become an indispensable foundation on which many researchers in academia and industry will rely.

New York University, New York, NY, USA
August 2021

Michael L. Pinedo

Preface

The topic of this book is decision-making at the interface of supply chain management and production scheduling. These are both very active research areas with profound practical impact through diverse applications. An essential component of such decision processes is coordination. In this book, coordination is studied in two different environments. We first study the coordination of multiple operational functions by a single centralized decision maker. This work generalizes the classical scheduling environment that typically addresses a single operation. We also study the coordination of the decisions of multiple decentralized decision makers who are self-interested but may be either cooperative or noncooperative. This work translates classical supply chain research into more detailed operational planning. The study of these topics within the book ranges broadly through the research areas of classical scheduling, combinatorial optimization, computational complexity, algorithm design, heuristic performance analysis, and cooperative and noncooperative game theory.

The book begins with two introductory chapters. Chapter 1 provides historical background on the research areas that support this work and identifies numerous applications that fall within our scope. Chapter 2 provides a concise overview of essential concepts and mathematical tools that are applied in the main body of the book.

The remainder of the book has two parts. Part I consists of Chaps. 3 through 6 on centralized supply chain scheduling problems, that is, those with a single decision maker who controls multiple operational functions. Chapter 3 studies offline versions of various integrated production and outbound distribution scheduling problems, whereas Chap. 4 is focused on online versions of such problems. Chapter 5 considers problems in which product pricing and production scheduling decisions are made together. Chapter 6 discusses situations where subcontracting and scheduling decisions are made jointly in order to optimize the overall system performance. In each chapter, we present solution algorithms and heuristics, analyze their performance, and evaluate the benefit of joint optimization.

Part II of this book consists of Chaps. 7 through 9. These chapters study decentralized problems, that is, those with multiple entirely or predominantly self-

interested players, in supply chain scheduling. Chapter 7 uses optimization models to study the cost of conflict that arises from unequal supply chain power and the benefit that results when players coordinate their schedules. Chapter 8 uses cooperative games to study the stability of coalitions between supply chain members and various schemes for compensating them using the gains from cooperation. Chapter 9 uses noncooperative games to study equilibrium solutions, their system performance, and possible ways to enhance that performance including by incentivizing the truthful reporting of private information.

Our motivation for writing this book arises in part from our consistent research interest in scheduling and related problems over more than 20 years. A second motivating factor is the influence of supply chain management which focuses on the coordination issue mentioned above, but typically does not relate it to the level of detailed scheduling decisions. We believe that the integration of these two research areas has both theoretical interest and practical value. In addition, because many scheduling problems have a simple structure that offers easily obtained insights, we have observed that such problems are used in numerous works to develop results within cooperative and noncooperative game theory. Those works naturally fall within the scope of our book.

Our writing process for the book has been an evolutionary one. It began with a broad concept of a supply chain, which includes, for example, a traditional vertical supply chain such as a supplier and a manufacturer, where a single decision maker coordinates multiple operational functions, and a horizontal supply chain such as multiple suppliers competing for the capacity of a manufacturer, where there are multiple decision makers at the same level. Additional generality was provided by the flexibility to model decision makers as either jobs that require scheduling or machines that schedule the jobs. The focus on intensified competition, which has been a major motivation for the development of supply chain management, has also motivated the many supply chain scheduling applications we discuss. We have been inspired by the growing variety of applications enabled by e-commerce that can be modeled as supply chain scheduling problems. A challenge that we have faced during our 4-year development of the book has been keeping up with the rapid pace of new research.

Scheduling research, which began more than 65 years ago, remains very active with many theoretical developments and innovative new applications appearing each year. Some trends that support increasing interest in this work in the future include the ever more important role scheduling plays in a supply chain, the widespread implementation of computer-based scheduling systems, and the increasing use of online platforms to schedule work among decentralized decision makers. Indeed, it is the potential growth of such applications that convinces us of the timeliness of our book project.

The intended audience of this book is researchers and graduate students who are interested in supply chain planning, extensions of classical scheduling, combinatorial optimization generally, and emerging applications of cooperative and noncooperative game theory. The book does not provide a comprehensive exposure to supply chain management, classical scheduling, combinatorial optimization, or

cooperative or noncooperative game theory. All these research topics are well supported by many excellent existing books, which can effectively prepare readers without previous background in these five topics. For students who are in possession of such background, this book could provide an extensive tour of those topics centered around the simple but highly intuitive paradigm of scheduling. We cite more than 400 references, which can be used as assigned readings in support of the content of the book. The book also contains more than 100 fully worked numerical examples that illustrate the broad range of mathematical techniques that model and solve supply chain scheduling applications.

Much of this book is based on work which we have done together or with other co-authors. For example, most of the content of Chap. 7 is based on four works which we originally wrote and which (based on the number of citations received) are viewed as foundational to the topic of supply chain scheduling. We have rewritten these works to make them more accessible and present the results in a unified way. However, we have also used parts of various works by other authors from the published open literature. In each case, we introduced some reorganization and rewriting, and of course appropriately referenced the work and where necessary (as advised by our publisher) obtained copyright permission. Under these guidelines, we have reproduced a number of proofs, especially mathematical proofs, without substantial changes. We have also contacted the most frequently cited authors to ensure that they believe that their work has been appropriately attributed. In several cases, they also suggested improvements to the presentation, which we appreciate.

This book has also benefited greatly from reading by several expert colleagues, including Alessandro Agnetis (Università di Siena), Zhixin Liu (University of Michigan-Dearborn), Daniel Ng (The Hong Kong Polytechnic University), Xiang-tong Qi (The Hong Kong University of Science and Technology), Bingling She (Imperial College London), George Vairaktarakis (Case Western Reserve University), Shuling Xu (Jiangnan University), and Zijin Zhang (University of Michigan). Their detailed and helpful comments are much appreciated, as are more general conversations with our many co-authors referenced within the book. We also thank our helpful and efficient editor at Springer, Jialin Yan. Finally, we thank our respective academic institutions for providing support for several summers, during which much of the writing of the book was completed.

College Park, MD, USA

Zhi-Long Chen

Columbus, OH, USA
August 2021

Nicholas G. Hall

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